BIOMECHANICS · EPHE 341

Sensors and Data Acquisition

How a physical quantity becomes data.

Lecture objectives

> An introduction to some biomechanics methodology. We will be using different equipment during this course, and the following is critical to appreciate how these devices quantify data.

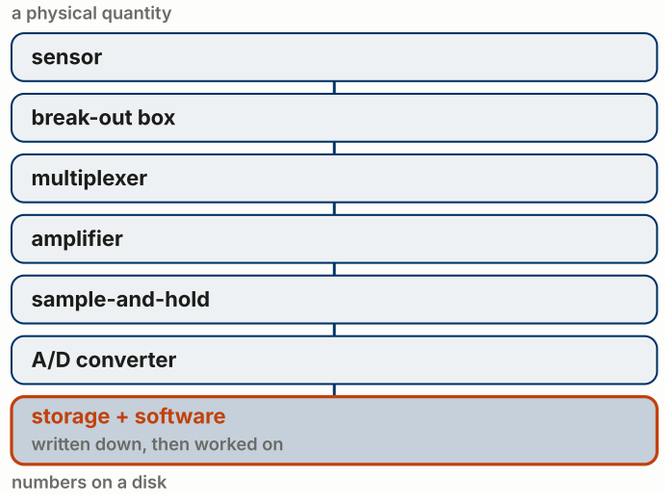
To learn about some basics of:

- **Sensors** used in biomechanics
- **Data acquisition**

Some important terms to look out for —

- Sensor
- Signal
- Analog
- Digital
- Sampling frequency (Hz)
- Calibration
- Data acquisition
- Break-out box
- Multiplexer
- Amplifier
- Sample-and-hold circuit
- A/D converter

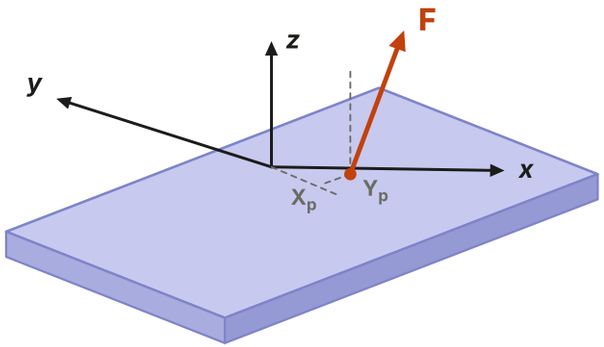
Everything in this lecture is one chain. A physical thing happens; a sensor turns it into a voltage; a run of electronics turns that voltage into numbers; software turns the numbers back into units you can analyze.



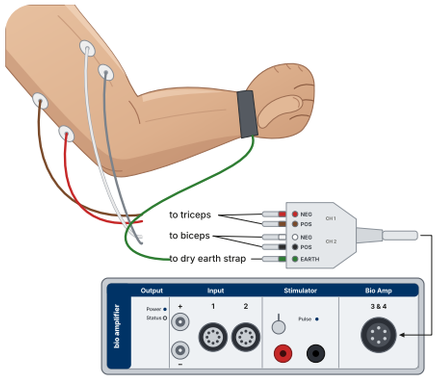
Sensors

sensor — a device that measures a physical quantity and converts it into a signal which can be read by an observer or by an instrument.

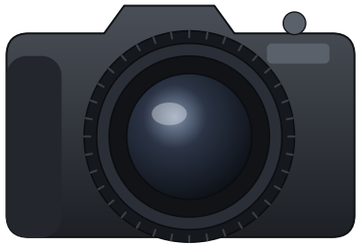
The signal to which most sensors convert the quantities they are designed to sense is **voltage**.



force plate — force to volts



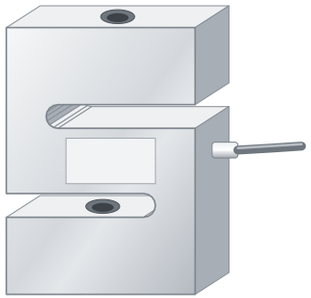
EMG — muscle activity to volts



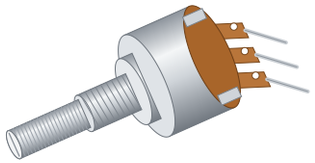
camera — light to pixels

Sensors

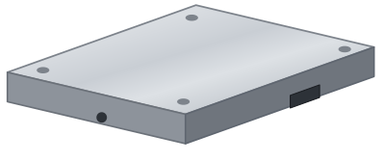
Some sensors commonly used in biomechanics — every one of them converts a physical value into an electronic signal, usually measured in **volts** (electrical potential).



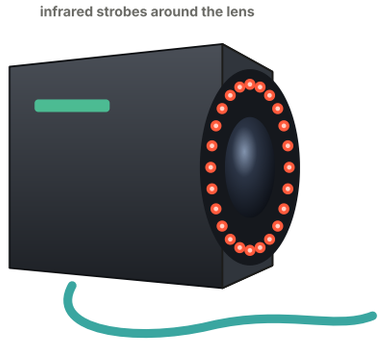
Strain gauge
converts force along one axis to volts



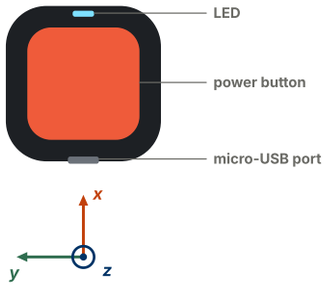
Potentiometer
converts angular displacement to volts



Force platform
converts force along three axes to volts



Motion capture
converts light to volts



IMU
converts accelerations to volts

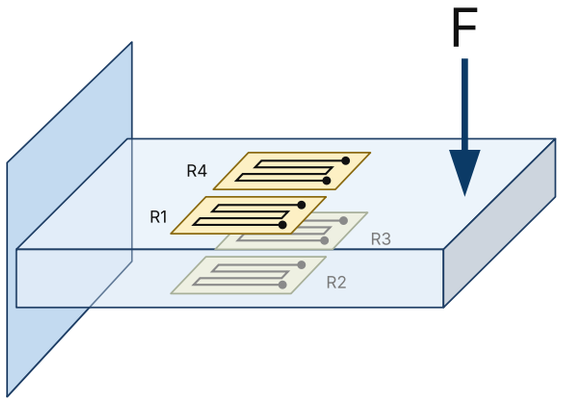
When necessary I may go into greater detail on how each sensor works — but for now all we need is that each one converts a physical value into an electronic signal.

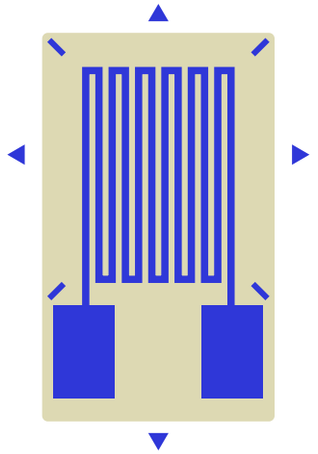
Sensor example: strain gauge

It may be useful to talk about how a **strain gauge** works, to see how a physical signal becomes an electrical potential.

A strain gauge is really a specially designed **resistive circuit** attached to a malleable (bendable) object.

Here we have four resistive circuits attached to a bendable beam, to which a force will be applied.





strain gauge resistor

Sensor example: strain gauge

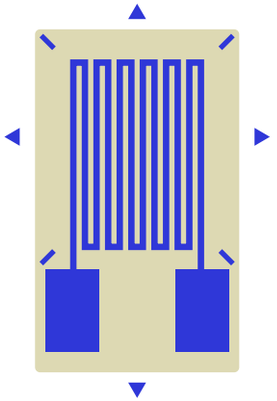
> A current is passed in one end of the foil pattern and out the other. The **resistance of that path** is what the gauge reports.

Electrical current in →

...around a long, thin, folded conductor...

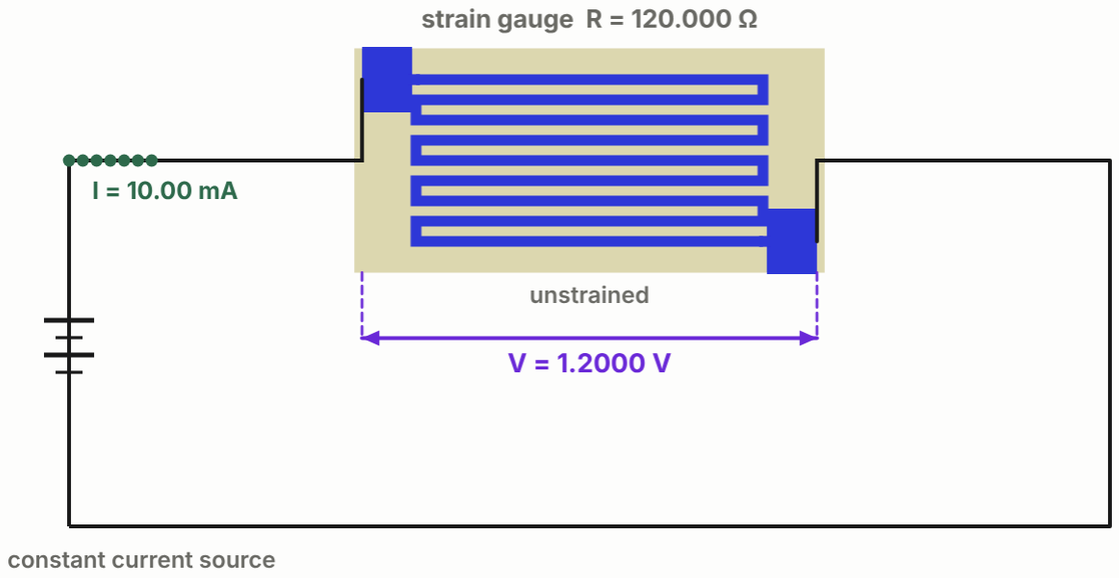
← Electrical current out

Stretch that conductor and it becomes longer and thinner, so the current has a harder time getting through: resistance goes **up**. Squash it and resistance goes **down**.



Sensor example: strain gauge

Current, voltage and resistance are tied together by **Ohm's law**: **V = I R**. The gauge is the R — so a change in strain becomes a change in voltage, and that is the entire sensor.



V = I R

the gauge is just a resistor

$$R = R_0(1 + GF \cdot \varepsilon)$$
$$R = 120(1 + 2.0 \cdot 0.000000)$$
$$R = \mathbf{120.000}\ \Omega\ (+0.000)$$

hold the current steady and the voltage follows R

$$V = I\,R = 0.010 \times 120.000$$
$$V = \mathbf{1.2000}\ V$$

R	I	V
120.00 Ω	**10.00 mA**	**1.200 V**

R = **120.000 Ω** · I = **10.00 mA** · V = **1.2000 V**

Driving a **constant current** through the gauge makes the voltage across it a direct read-out of its resistance — and therefore of the strain. That is why a

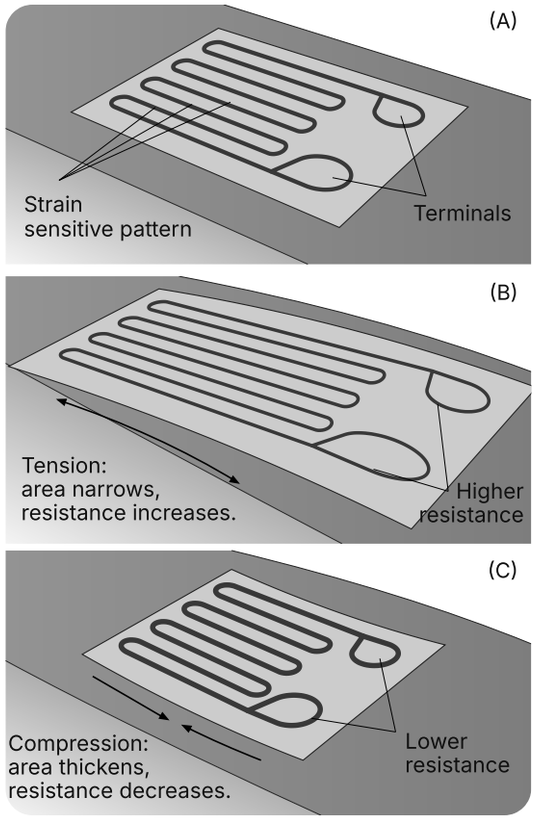
THE SOURCE HOLDS

Sensor example: strain gauge

When the strain gauge resistor is attached to the physical object and the object is **bent**, the resistor bends with it.

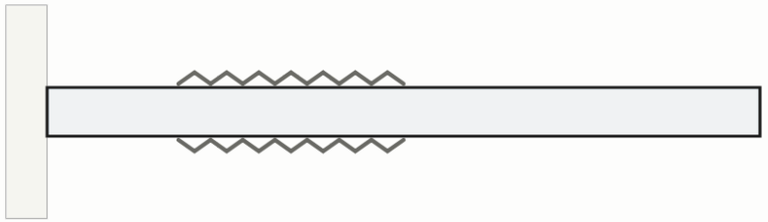
This allows the bend of the physical object to be related to a physical **voltage**.

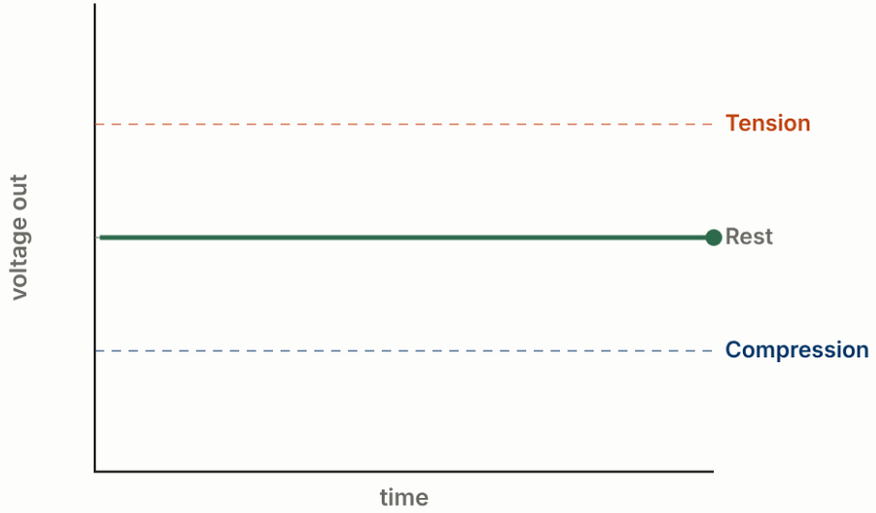
As seen in **B**, if the gauge is in **tension** the electrical resistance to flow **increases**. In **C**, if the gauge is in **compression** the resistance **decreases**.



Sensor example: strain gauge

We could measure the voltage in conditions where the beam was **compressed**, in **tension**, or **at rest** — and we could watch it change as the force does.





the top gauge is at rest

strain in the gauge

$\varepsilon = 0\ \mu\varepsilon$

resistance follows the strain

$\Delta R = GF \cdot \varepsilon \cdot R = 2.0 \cdot \varepsilon \cdot 120\ \Omega$

$R = 120.000\ \Omega$ (+0.000)

and V = IR turns that into volts

$V_{out} = 0.0\ mV$

F	R	V
0 N	**120.00**	**0**

force **0** N · strain **0** με · resistance **120.000** Ω · out **0.0** mV

a gauge is a resistor glued to something bendy — bend it and its resistance changes, and V = IR turns that into a voltage you can record

Applied force **0 N**

Signals

> **signal** — a physical quantity or effect that can be varied in such a way as to convey information. It can be either **analog** or **digital** (continuous or discrete, respectively).

> An **analog clock** displays the time continuously; a **digital clock** displays only discrete points in time.

The resolution of the digital clock describes the interval between those points — the higher the resolution, the smaller the interval.

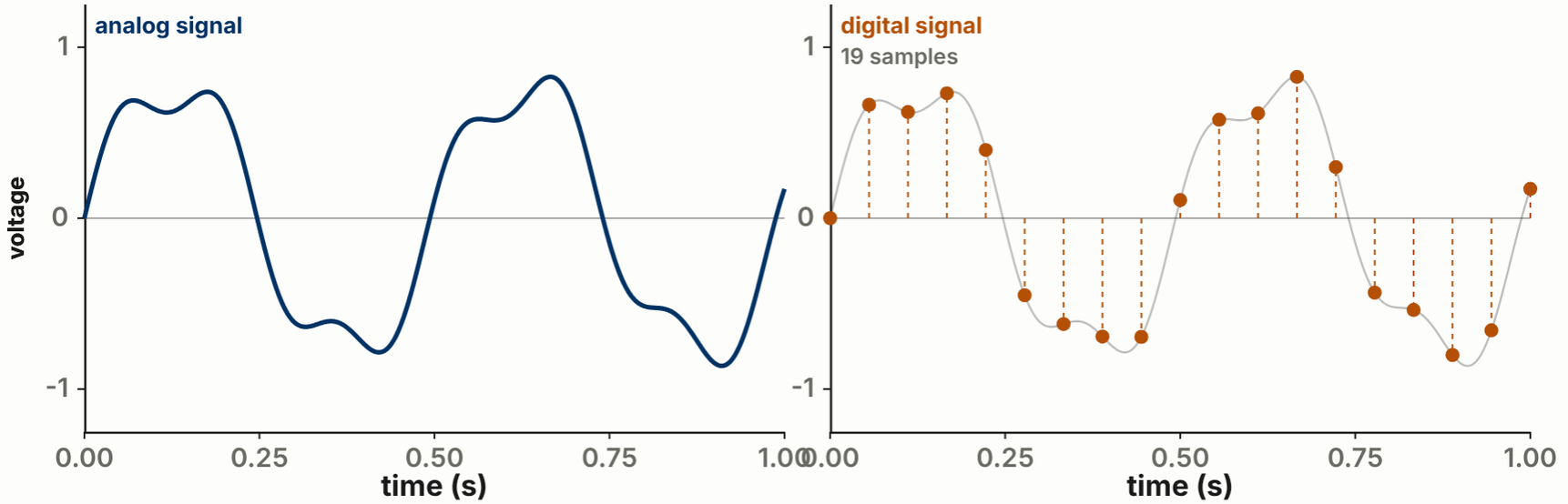


analog clock

Signals

If a signal is **analog** it means there are no separations, based on time, between points in the signal.

If a signal is **digital** it means there **are** separations between points in the signal.



signal **2.0** Hz · sampled at **18** Hz · **56** ms between samples

9.0 samples per cycle

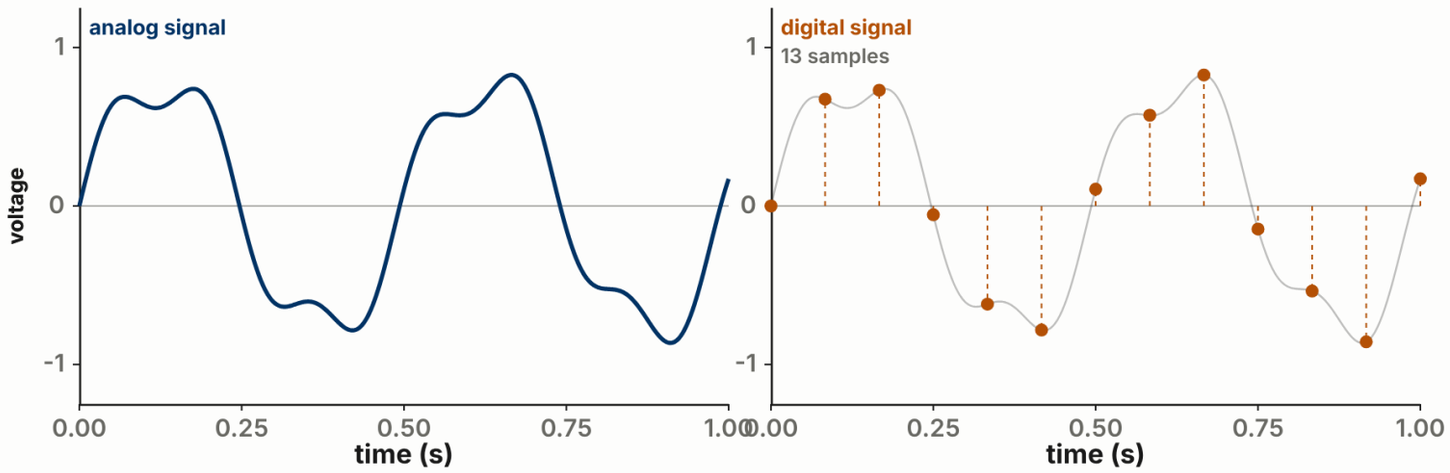
SHOW **both** analog digital **Join the dots**

Signal frequency **2.0 Hz**

Use your eyes as an example. Viewing an activity without blinking gives you a continuous, connected representation. Now close and open your eyes every second, for a very short time — you are only able to see **samples** of the activity.

Sampling

sampling — the process of converting a continuous signal into a digital signal (a numeric sequence). That is, “taking samples”.



signal **2.0** Hz · sampled at **12** Hz · **83** ms between samples

6.0 samples per cycle

SHOW **both** analog digital **Join the dots**

Sampling rate **12 Hz**

Signal frequency **2.0 Hz**

Most physical signals are inherently **continuous**, but when we use sensors to quantify the data we must convert them to a discrete digital signal by taking discrete samples.

The **sampling frequency** is defined as the number of samples in one second, and is expressed in **hertz (Hz)**.

12 Hz means twelve numbers per second, and 83 ms of the world that nobody wrote down between each pair of them.

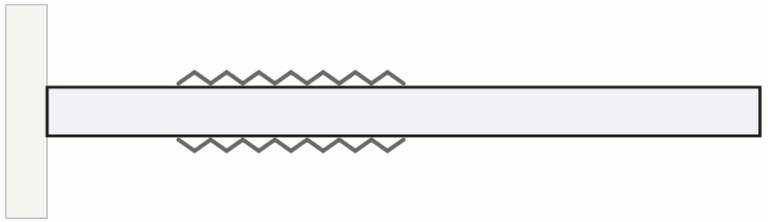
Calibration

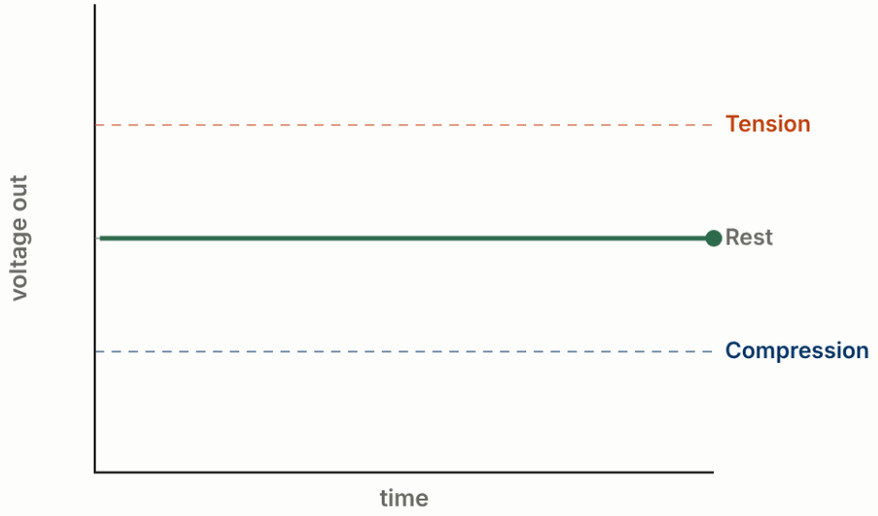
calibration — the validation of specific measurement techniques and equipment: a comparison between measurements, one of known magnitude and another of unknown magnitude.

For a sensor's output (usually voltage) to yield meaningful information, it must be calibrated to a **known magnitude** of the quantity being measured.

Using our strain gauge example again: when we use the sensor we have a value in **volts** — but what we really want is a measure in **newtons**. Nothing inside the instrument knows what a newton is.

$$1\ \text{volt} = ?\ \text{newtons}$$





the top gauge is at rest

strain in the gauge

$\varepsilon = 0\ \mu\varepsilon$

resistance follows the strain

$\Delta R = GF \cdot \varepsilon \cdot R = 2.0 \cdot \varepsilon \cdot 120\ \Omega$

$R = 120.000\ \Omega\ (+0.000)$

and V = IR turns that into volts

$V_{out} = 0.0\ \text{mV}$

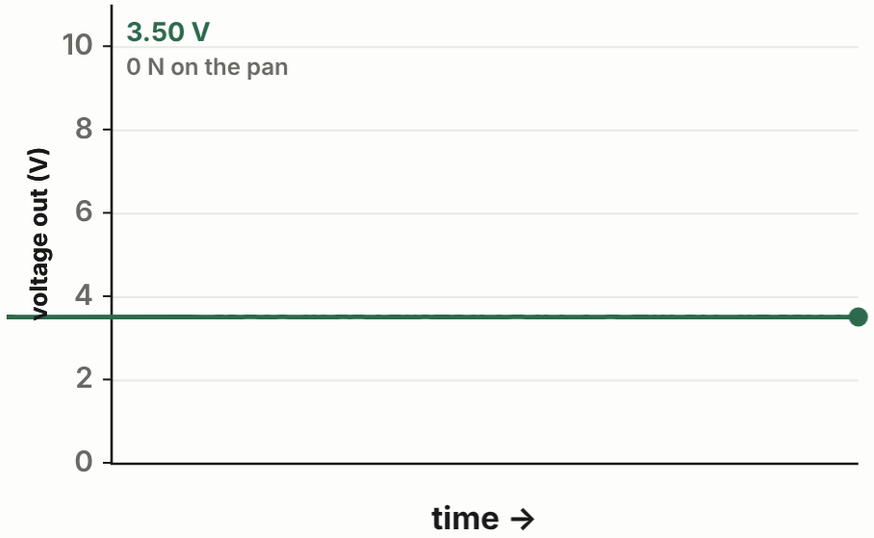
Sensor calibration

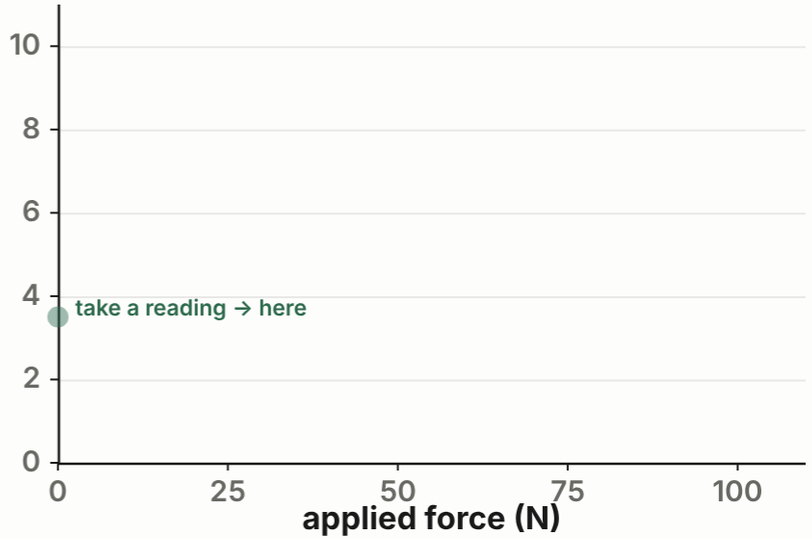
How is a sensor calibrated? — in this example, a force transducer

We could take the force transducer we used earlier and apply various **known forces** to it. Then we graph the value of the force against the voltage the sensor outputs.

If the relationship is **linear**, we can fit a linear equation to determine the force for any measured voltage. **y** is the voltage, **m** the slope, **x** the force and **b** the offset.

$$y = mx + b$$





no readings yet

Hang a known weight on the sensor, wait for the trace to settle, then **take a reading**. Two readings give a line; three let you check it.

not calibrated yet — **0** of 2 readings needed

the sensor only ever gives volts. The calibration is the thing that turns volts into newtons, and it comes from weights you already knew the answer to.

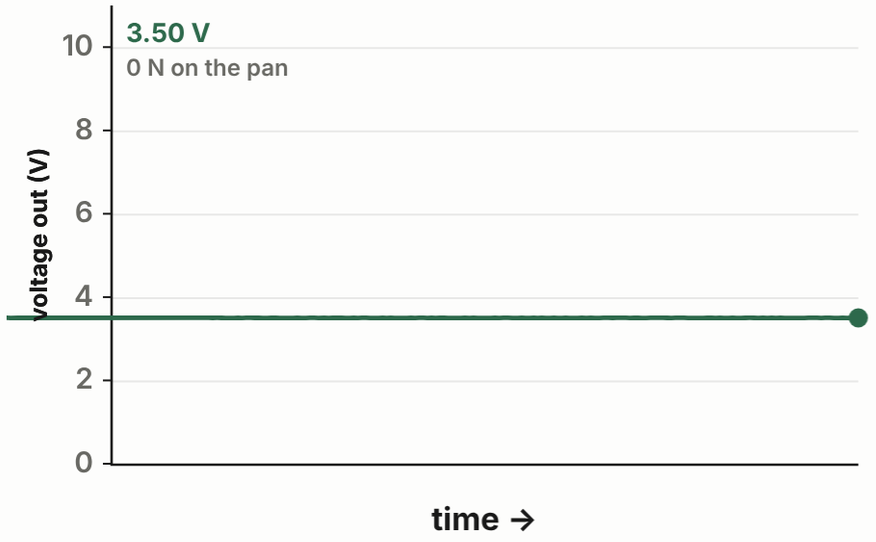
Sensor calibration

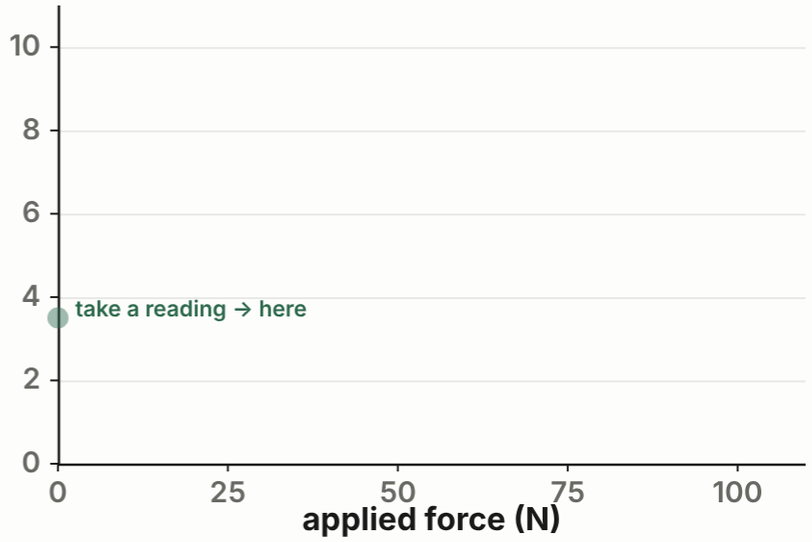
For example

> Below are the applied force and voltage output pairs for a given sensor. **Determine the slope and the offset.**

Applied force (N)	Voltage output (V)
25	5.0
50	6.5
75	8.0

Hang each weight in turn, wait for the trace to settle and take a reading — the three rows below should come out as the three rows above.





no readings yet

> Hang a known weight on the sensor, wait for the trace to settle, then **take a reading**. Two readings give a line; three let you check it.

Sensor calibration

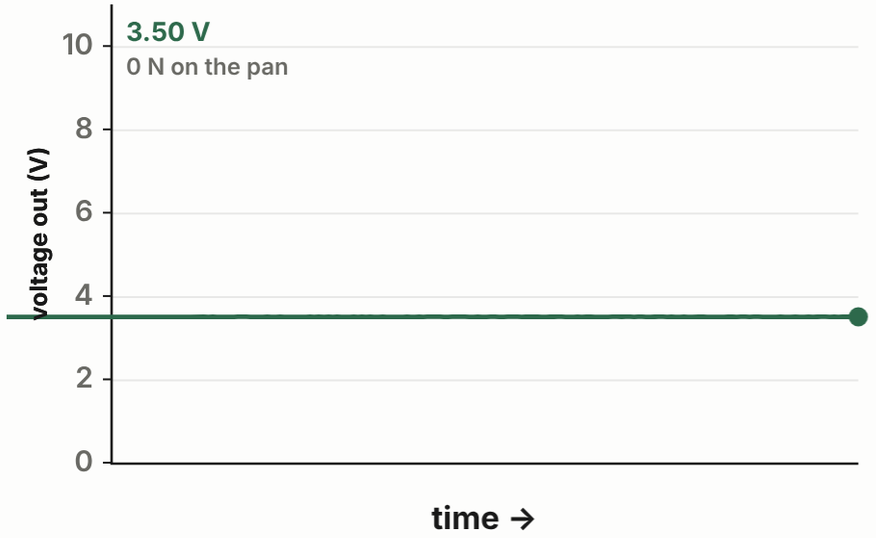
Solution — the slope

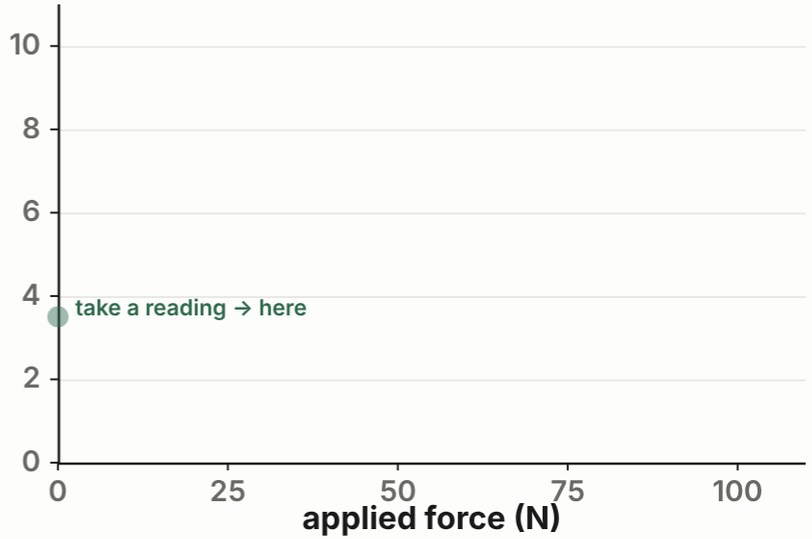
By plotting the given data we can see the relationship is **linear**. The slope is the ratio of **rise to run**; the **offset** then comes from plugging a known pair into the equation for a line.

$$\text{slope} = \frac{\Delta y}{\Delta x} = \frac{6.5 - 5.0}{50 - 25} = \mathbf{0.06\ V/N}$$

$$y = mx + b \rightarrow 5 = (0.06)(25) + b$$

$$\mathbf{b = 3.5\ V}$$





no readings yet

Hang a known weight on the sensor, wait for the trace to settle, then **take a reading**. Two readings give a line; three let you check it.

not calibrated yet — **0** of 2 readings needed

Sensor calibration

Now use the sensor

> With the slope and the offset we can determine the force for any voltage output — the sensor is said to be **calibrated**. Plug a measured voltage into the equation for a line, using our calculated slope and offset.

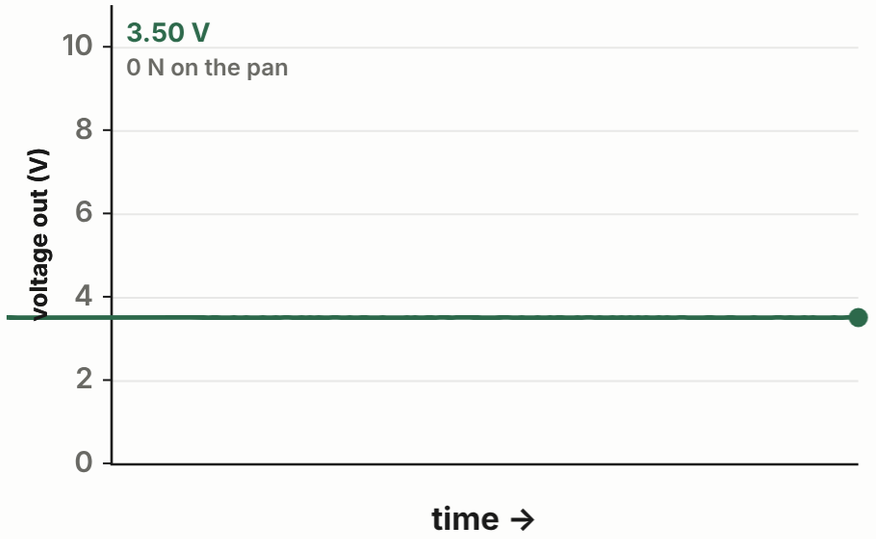
> Record the three points first, then slide the voltage. The green lines are the arithmetic on the right, drawn.

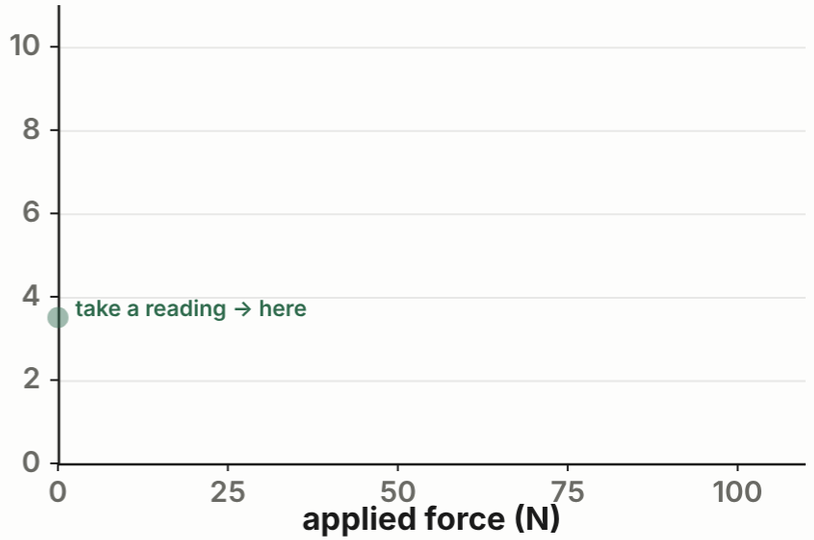
e.g. for an output of 6 volts, the force would be:

$$y = mx + b$$

$$6\ \text{V} = (0.06\ \text{V/N})\ x + 3.5\ \text{V}$$

$$x = 41.6\ \text{N}$$





no readings yet

> Hang a known weight on the sensor, wait for the trace to settle, then **take a reading**. Two readings give a line; three let you check it.

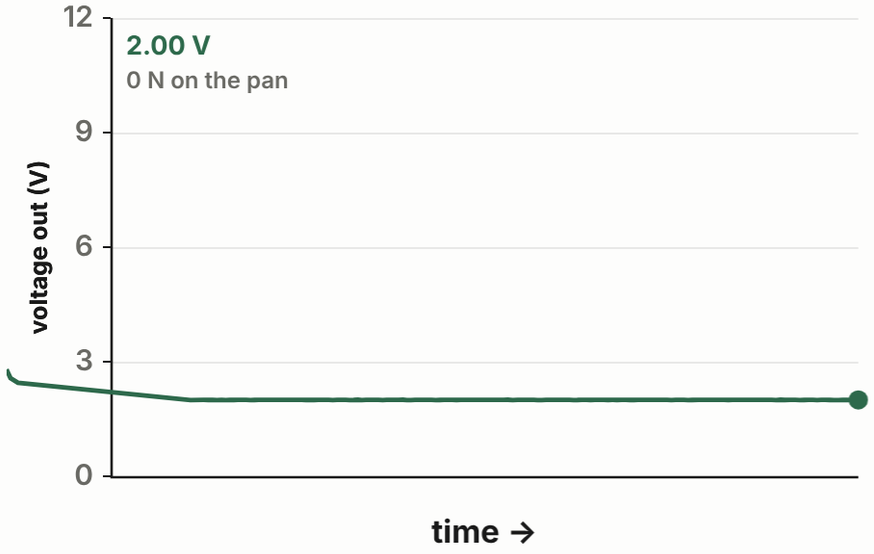
Sensor calibration

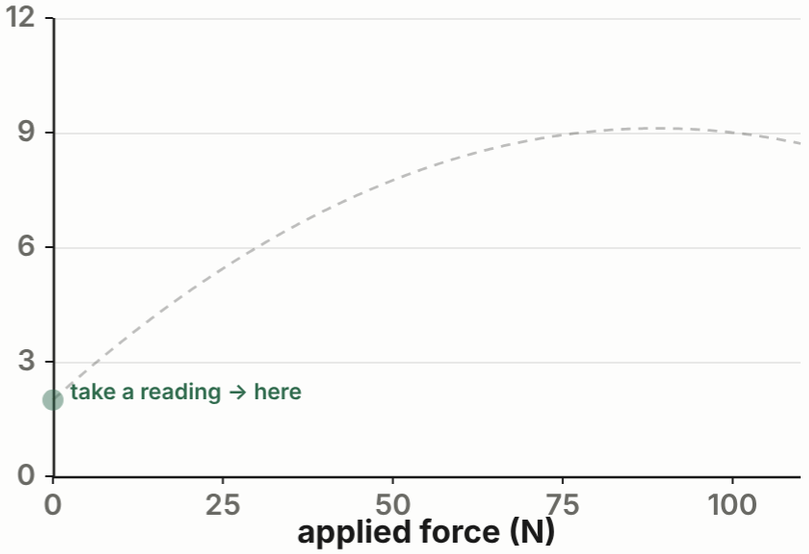
When the relationship is not a straight line

If the relationship between applied force and voltage output is **non-linear**, the data are fitted to a **polynomial** equation.

In such cases we need more than just the slope and offset — we need the **coefficients** that describe the polynomial. This calibration equation can then be used to convert from a measured voltage to a force in newtons.

$$y = -0.1x^2 + 1.2x$$





no readings yet

Hang a known weight on the sensor, wait for the trace to settle, then **take a reading**. Two readings give a line; three let you check it.

not calibrated yet — **0** of 2 readings needed

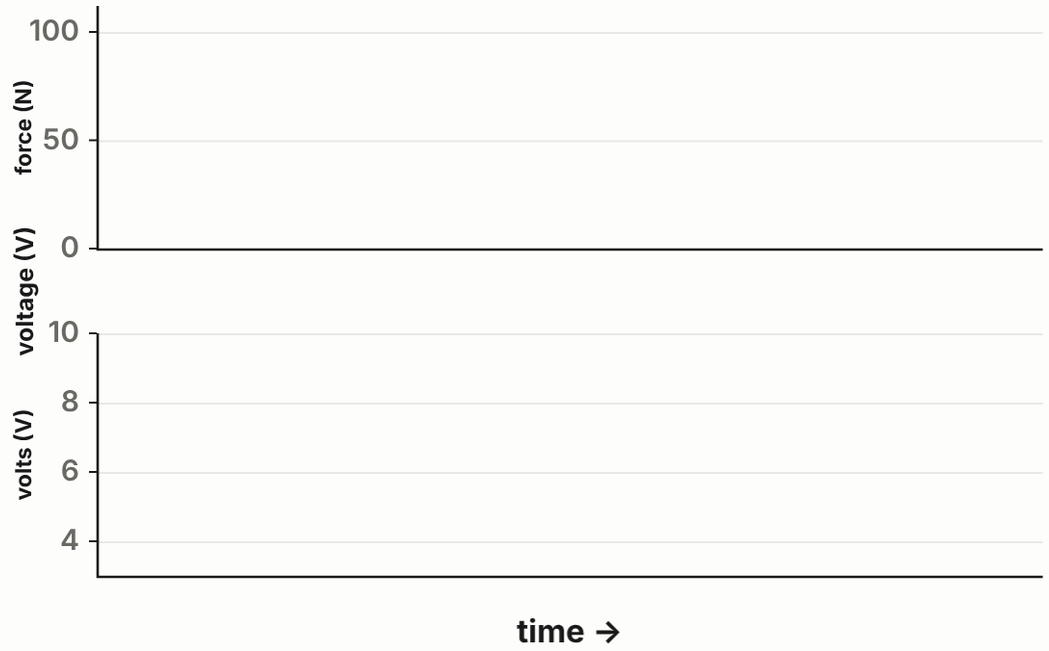
Sensor calibration

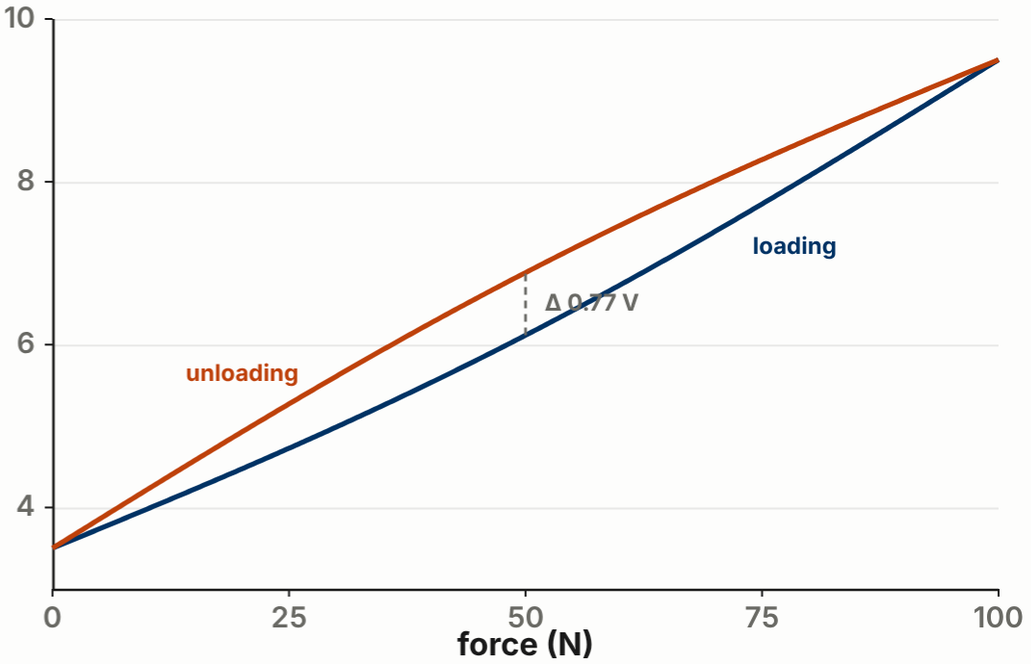
Hysteresis

> The influence of the **previous history** of a system on its subsequent response to a given stimulus.

Hysteresis occurs when a **different force-to-voltage relationship** exists when the force is incrementally increased, compared with when it is incrementally decreased.

Hysteresis is undesirable because it would require different calibration coefficients — or different equations — for **loading** and **unloading**.





Sensor calibration

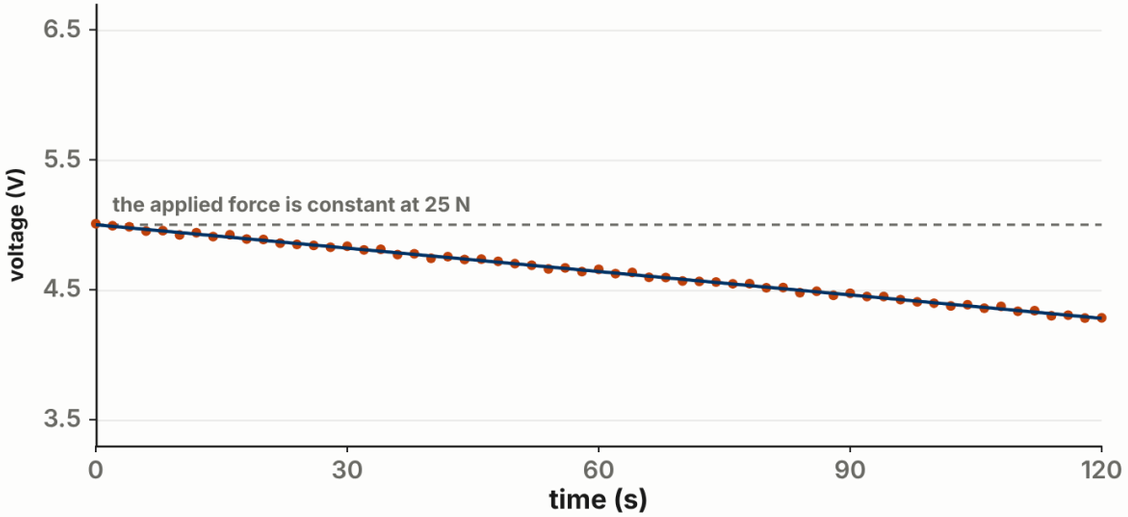
Drift

> A gradual change in a transducer's voltage output **without an accompanying change in the applied force**.

Drift can occur as a result of the transducer warming up, or as an effect of continued use.

If the drift is **linear**, the slope can be determined and the voltage output can be adjusted.

Throughout the figure the applied force is **constant**.



the output slides by **-0.72** V over two minutes

put those volts through the calibration and the force appears to go from **25.0 N** to **13.0 N**, without anybody touching the sensor

Correct the drift

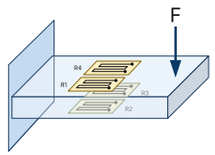
Drift rate **-6 mV/s**

Data acquisition

How are data collected from a sensor?

a physical quantity → **sensor** → break-out box → multiplexer → amplifier → sample-and-hold → A/D converter → storage + software → numbers on a disk

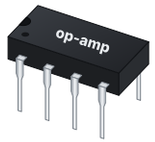
turns a physical thing into volts



1. sensing unit



2. power supply



3. op-amp

A sensor normally includes:

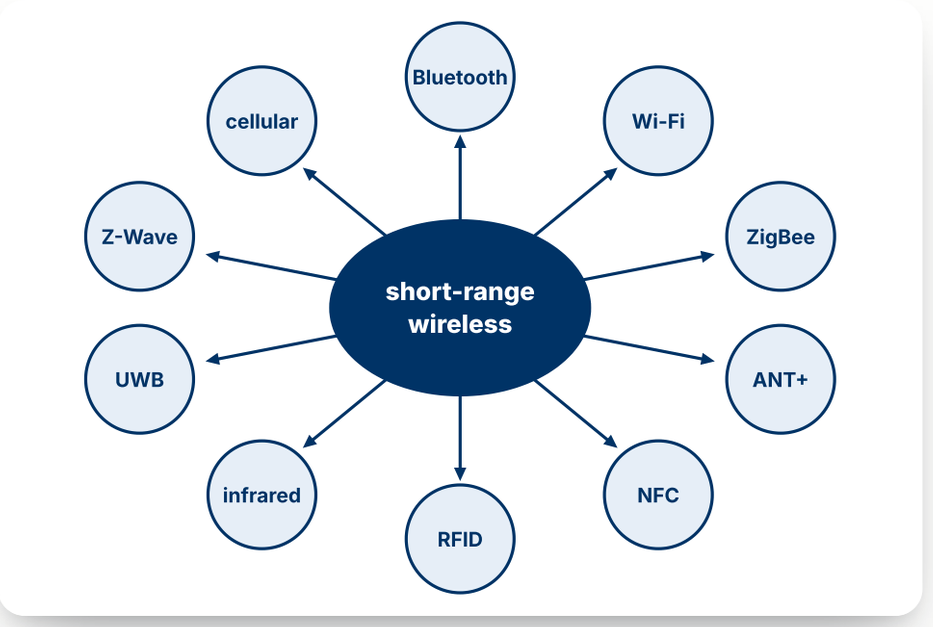
- The **physical sensing unit** — e.g. a strain gauge
- A **power supply** — batteries or plug-in
- An **operational amplifier** — a dedicated circuit that amplifies the voltage from the sensor and can perform operations on the signal, from simple to complex maths

Data acquisition

How are data collected from a sensor? — wireless

In biomechanics it is important to consider how we collect data and store it in the computer.

A wireless sensor does the first few boxes of the chain on board and sends numbers, not volts. Convenient — but the sampling rate, the range and the resolution were all decided by somebody else, before you switched it on.

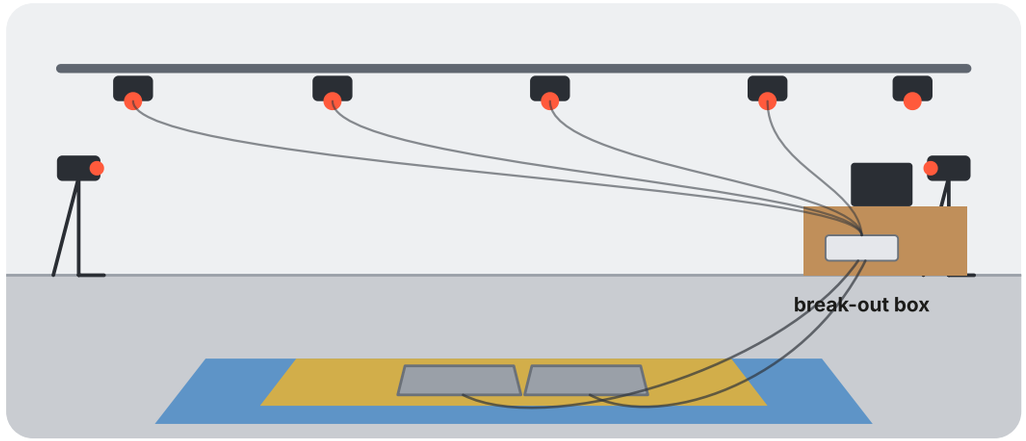


Data acquisition

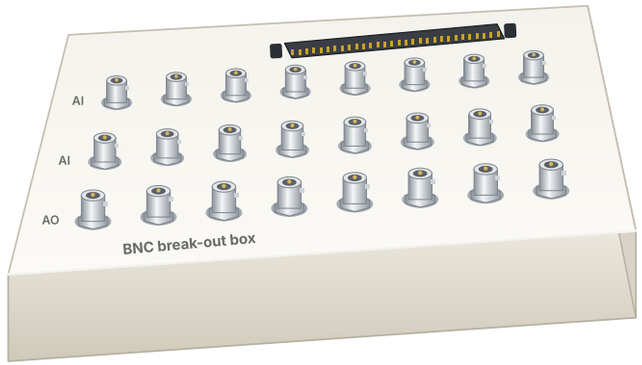
How are data collected from a sensor? — wired



> Each sensor gets a terminal on a **break-out box**, and everything leaves on one cable.



a wired lab — every one of those cables ends up in a break-out box

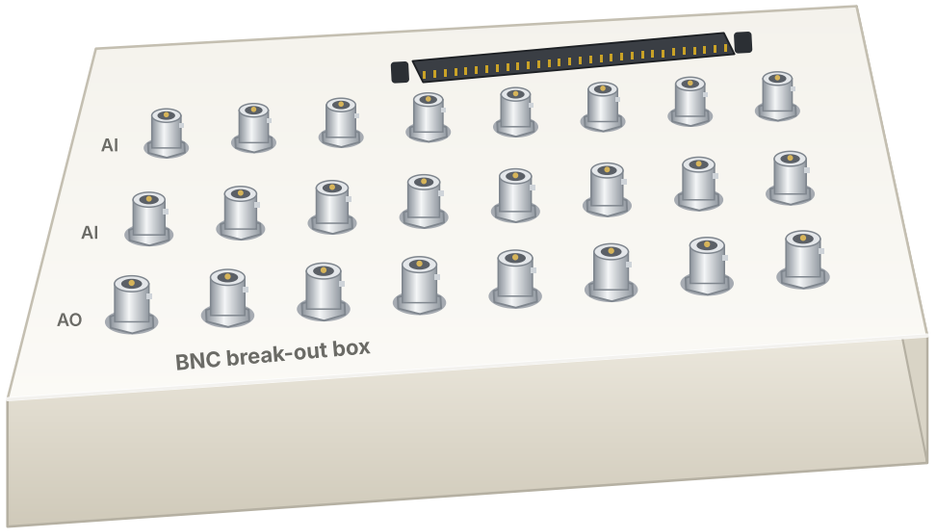


the break-out box itself — one terminal per sensor

Data acquisition

Break-out box

- A plastic or metal box with one or more rows of **terminals** (or connectors).
- A wire for each sensor connected to the break-out box is bundled into one large cable that connects to the rest of the **data acquisition (DAQ)** system.
- Can be built into a DAQ system, or it can be a stand-alone unit.
- Provides a means of attaching the sensors to the DAQ system **without soldering**.



Data acquisition

How are data collected from a sensor?



> In most cases we need to record from more than one sensor at a time. However, **DAQ units are not good at multitasking**.

> We could use one DAQ system for each signal being recorded — but there is a less expensive alternative: **multiplexing**.

Each connection to a multiplexer is called a **channel**.

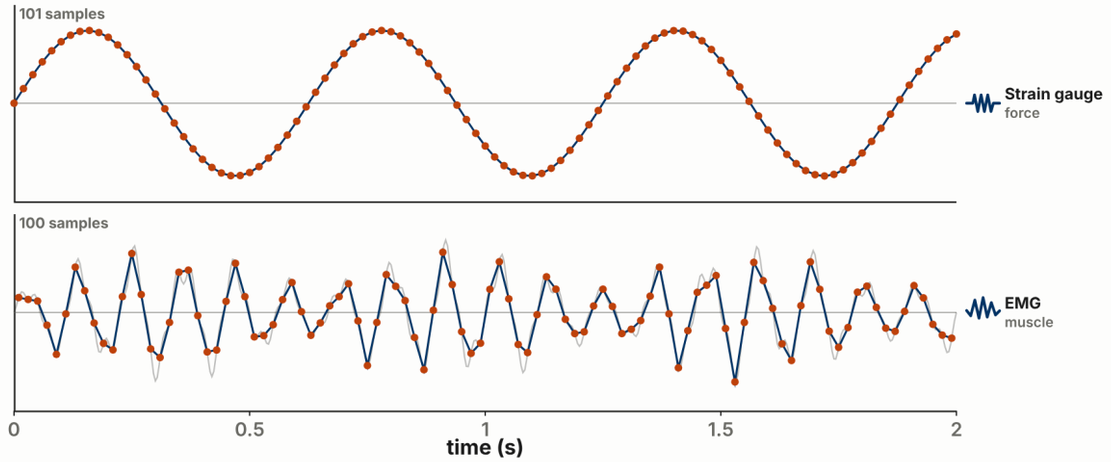
Data acquisition

Multiplexer

The DAQ draws the first sample from the first sensor; the multiplexer then switches the circuit to receive a sample from the second sensor; once all connected sensors have been read, the multiplexer switches back to the first.

> One problem with multiplexing is that the **effective sampling rate is decreased** — by a factor of the number of channels being multiplexed.

For example, consider a DAQ with the ability to draw 100 samples per second (100 Hz). If there are **8 sensors** connected, the multiplexer will switch between the 8 channels, giving an actual sampling rate of



2 sensors on one **100 Hz** DAQ · each channel actually gets **50.0 Hz** · last channel lags the first by **10.0** ms

the multiplexer reads one channel at a time and then starts again — so the rate is shared out, and the samples in a set are not simultaneous

CHANNELS | − sensor | + sensor

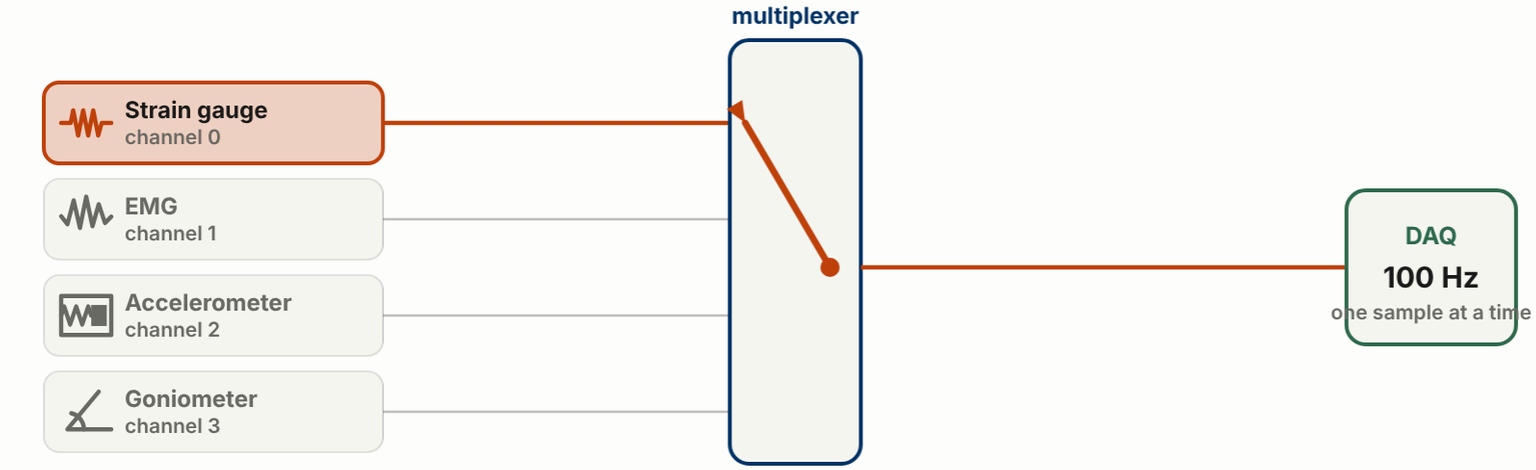
Skew shown

DAQ rate **100 Hz**

Data acquisition

Multiplexer — one channel at a time

> Each connection to a multiplexer is called a **channel**. The switch can only look at one of them at a time: it points at a channel, takes a sample, and moves on.



what the DAQ actually records — one channel per sample, in turn

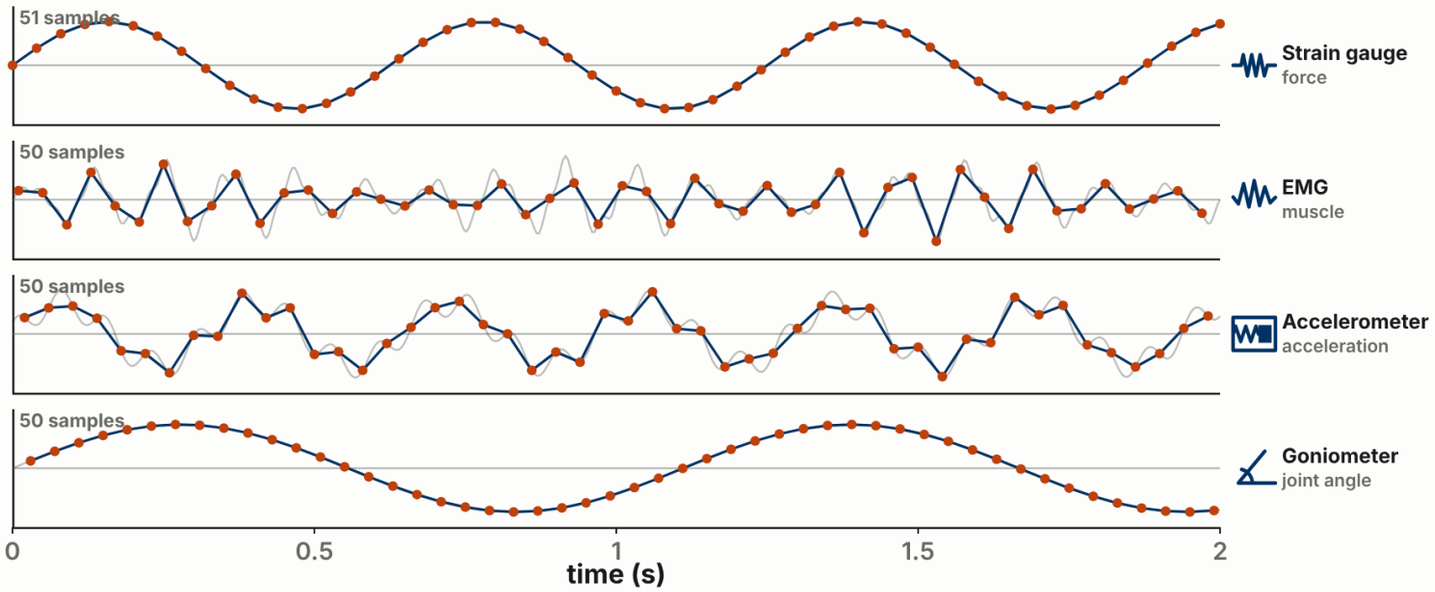
the arm is on **channel 3** (Goniometer) · it stays for **10.0 ms**, then moves on

The DAQ runs at **100 Hz**, but it has to share that between **4** channels, so each sensor is really sampled at **25.0 Hz**. And because the arm visits them one after

CHANNELS | – channel | + channel

Data acquisition

Multiplexer — sampling skew



4 sensors on one **100 Hz** DAQ · each channel actually gets **25.0 Hz** · last channel lags the first by **30.0** ms

the multiplexer reads one channel at a time and then starts again — so the rate is shared out, and the samples in a set are not simultaneous

CHANNELS | – sensor | + sensor | **Skew shown**

DAQ rate — **100 Hz**

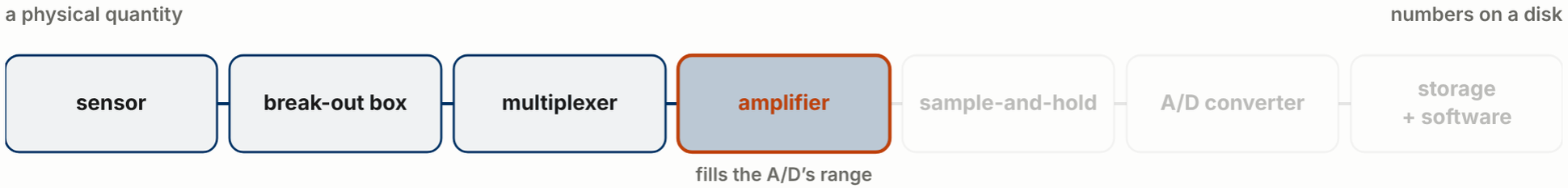
A second problem with multiplexing is that, since the samples are drawn **sequentially**, there is a delay between each one: the 8 samples are **not drawn simultaneously**.

This is called **sampling skew**, and it increases with the number of channels being sampled.

Toggle “Skew shown” to see it. With skew hidden, every channel appears to be sampled at the same instants — which is the comfortable assumption most analysis software quietly makes.

Data acquisition

How are data collected from a sensor?



> Voltages generated by sensors of physiological parameters are typically **very small** in amplitude — sometimes even after initial amplification.

> The number of times that a signal's amplitude is increased is called a **gain**.

DAQ systems usually come with preset choices of gain — e.g. **×1, ×10, ×100**.

Noise is often introduced through the wires that bring the signal to the amplifier; a **preliminary amplifier** located close to the sensor is therefore often used.

Data acquisition

How are data collected from a sensor?



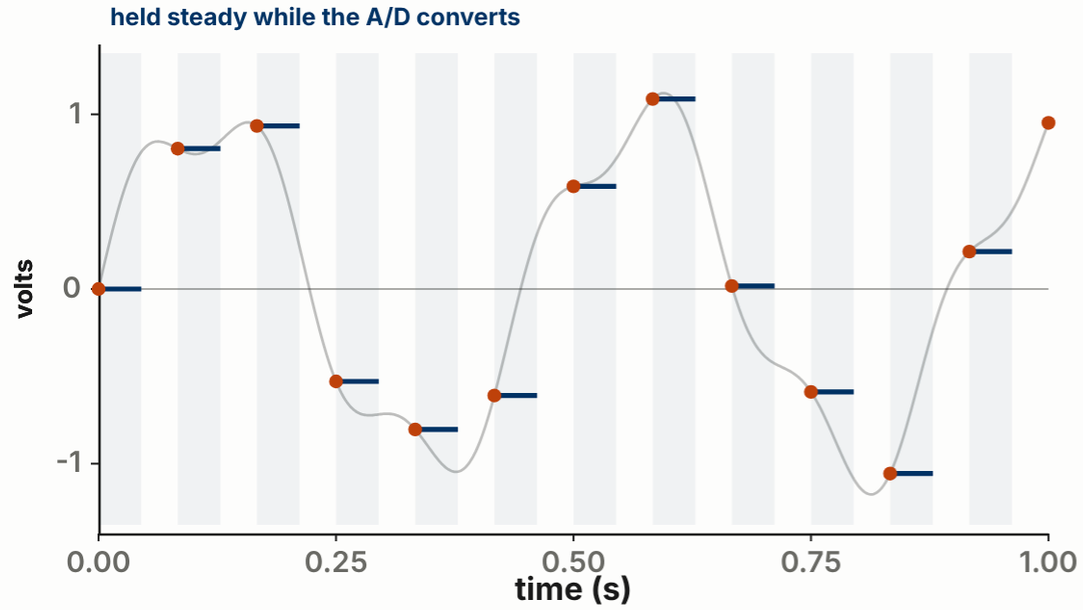
> Sample-and-hold circuit — **it takes a picture.**

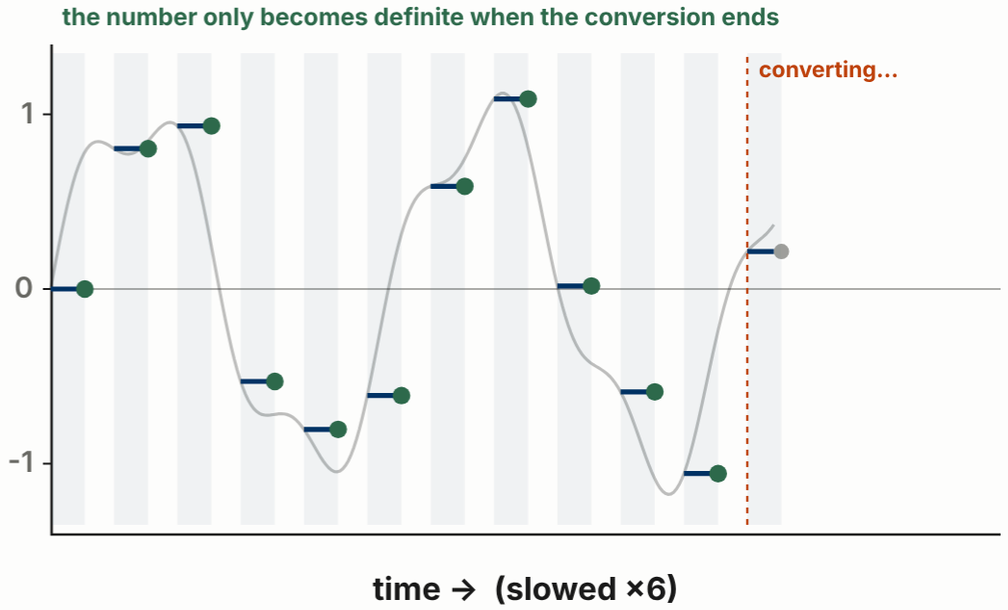
The analog-to-digital converter works to divide the analog signal into discrete values: one value at each discrete time point.

The conversion of a given voltage into a discrete value **takes a certain amount of time**, during which the A/D tries to determine the number that most closely corresponds to the applied voltage.

Sample-and-hold circuit

To convert accurately, the voltage input **must not change much** — or the number produced by the A/D will not be an accurate reflection of the voltage.





conversion takes **45** ms · sampling at **12** Hz

the sample-and-hold freezes the voltage the instant the sample is taken, and keeps it there until the A/D has finished deciding on a number — watch a dot on the right fade up as its converter works, and land solid only when it is done

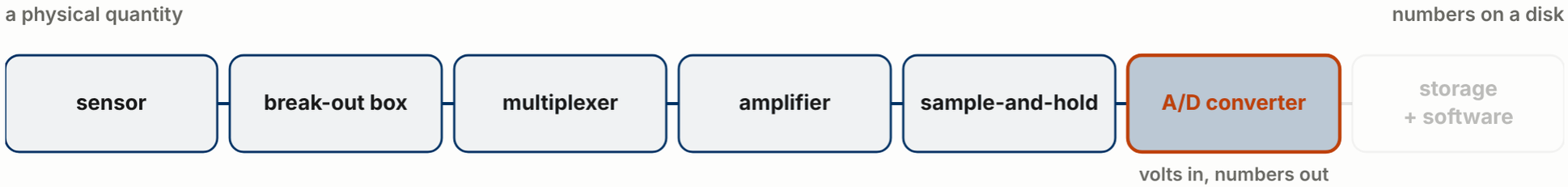
Sample-and-hold ON **Restart the run**

Conversion time **45 ms**

Sampling rate **12 Hz**

Data acquisition

How are data collected from a sensor?



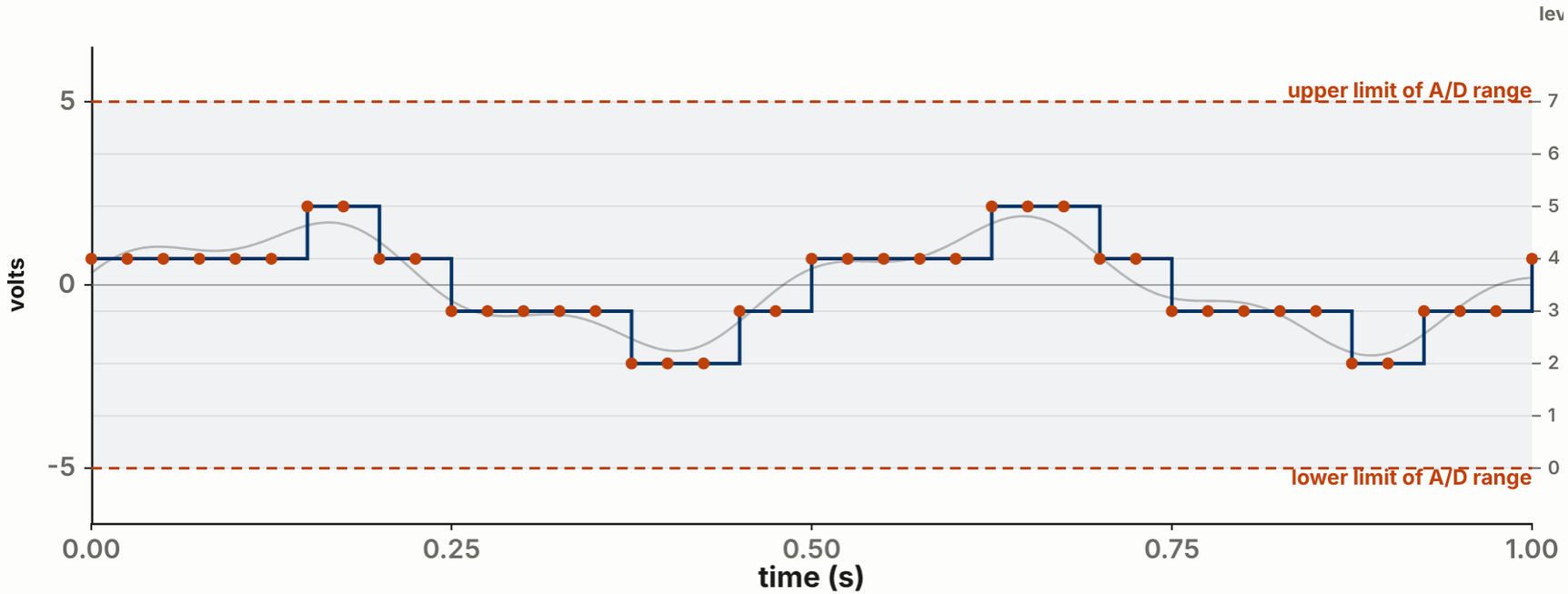
> The **analog-to-digital converter** makes up the most important component of a DAQ system. Its purpose is to turn a continuous, analog signal into a series of **discrete numbers**.

A/D converters are limited by three things:

1. The **range of voltages** that they can read — window and amplification
2. The number of **magnitude options** — magnitude resolution, or steps
3. The number of **time options** — temporal resolution

Analog-to-digital converter

All three limits, in one figure. Change one at a time and watch what the stored signal loses.



range **-5** to **5** V · **3** bits = **8** levels, one step is **1.429** V · **40** Hz

only 8 levels to choose from, so the staircase is coarse — this is magnitude resolution

A/D RANGE **−5 to +5 V** | −10 to +10 V | 0 to 1 V

Magnitude — bits

3 bit → 8 levels

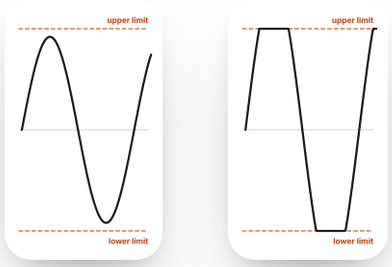
Analog-to-digital converter

1. The range of voltages it can read

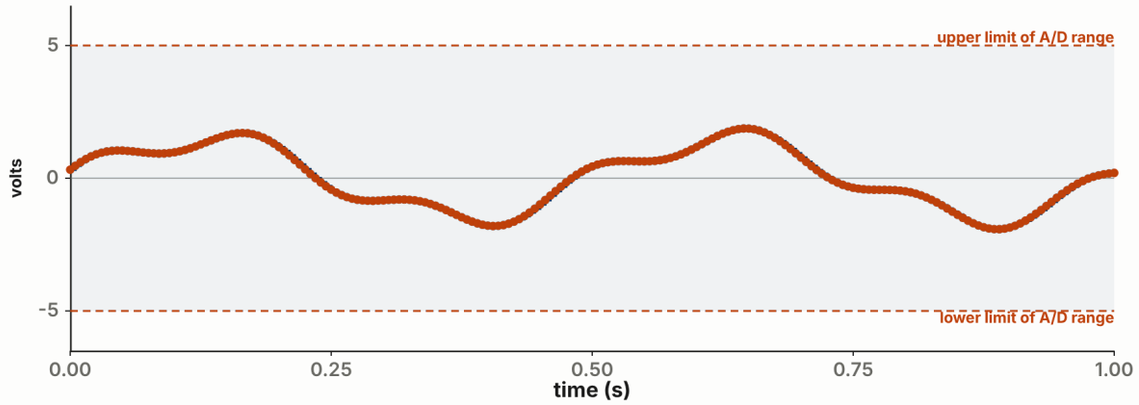
A/Ds can operate over only a **specific range of voltages** — e.g. 0 to 10 V, or −5 to +5 V.

If a voltage is applied beyond the upper or lower limits, the digitized signal will be truncated, or **‘clipped’**.

This is why it is important to know how to **amplify the signal**.



within range clipped



range **-5** to **5** V · **12** bits = **4096** levels, one step is **0.00244** V · **200** Hz

the signal fills the window, the steps are fine and the rate keeps up

Amplifier gain ×1

Analog-to-digital converter

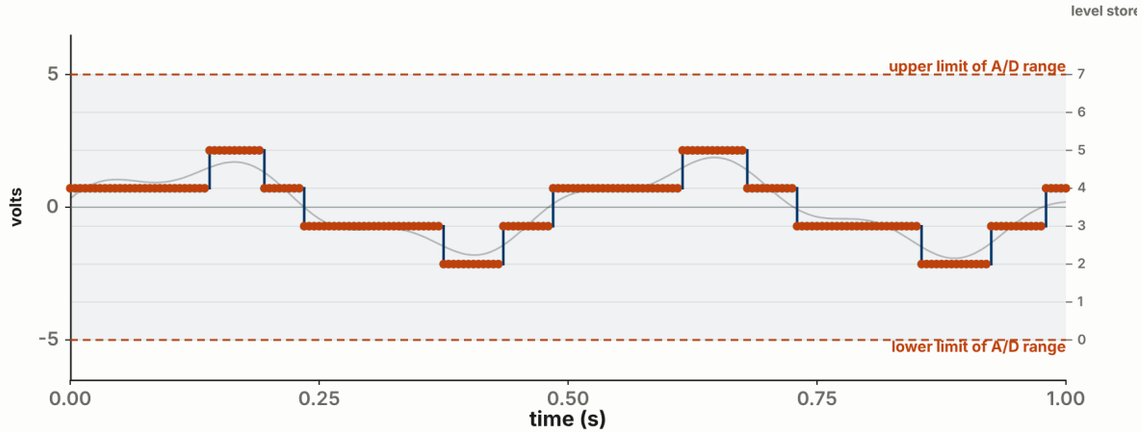
2. The number of magnitude options

The magnitude resolution of an A/D is usually expressed in units of **bits**.

‘Bits’ is short for ‘binary digits’, and refers to information made up of digits that can exist in only one of two states — **0 or 1**.

An 8-bit A/D has 256 ($2^8 = 256$) levels, meaning that 8 digits, or place-values, of either 1 or 0 are required to represent 256 individual numbers.

Bits		Number of levels
8	$2^8 =$	256
12	$2^{12} =$	4 096
16	$2^{16} =$	65 536



range **-5** to **5** V · **3** bits = **8** levels, one step is **1.429** V · **200** Hz

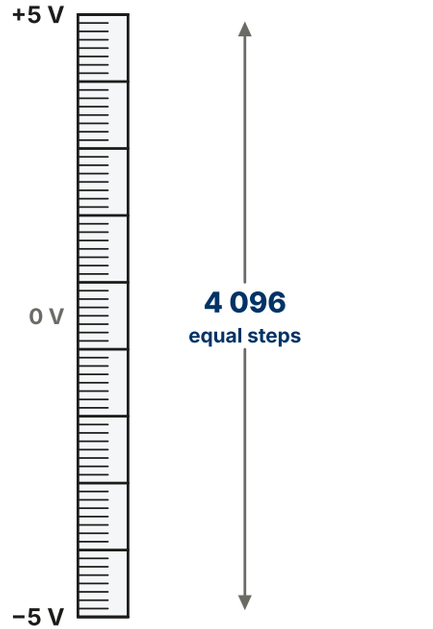
only 8 levels to choose from, so the staircase is coarse — this is magnitude resolution

Magnitude — bits

3 bit → 8 levels

Analog-to-digital converter

Resolution — the size of one step



(the ticks are fewer than 4 096 so they can be seen)

An A/D's range is divided into a specific number of **equal steps**. The number of steps that divides the range is called the **resolution**.

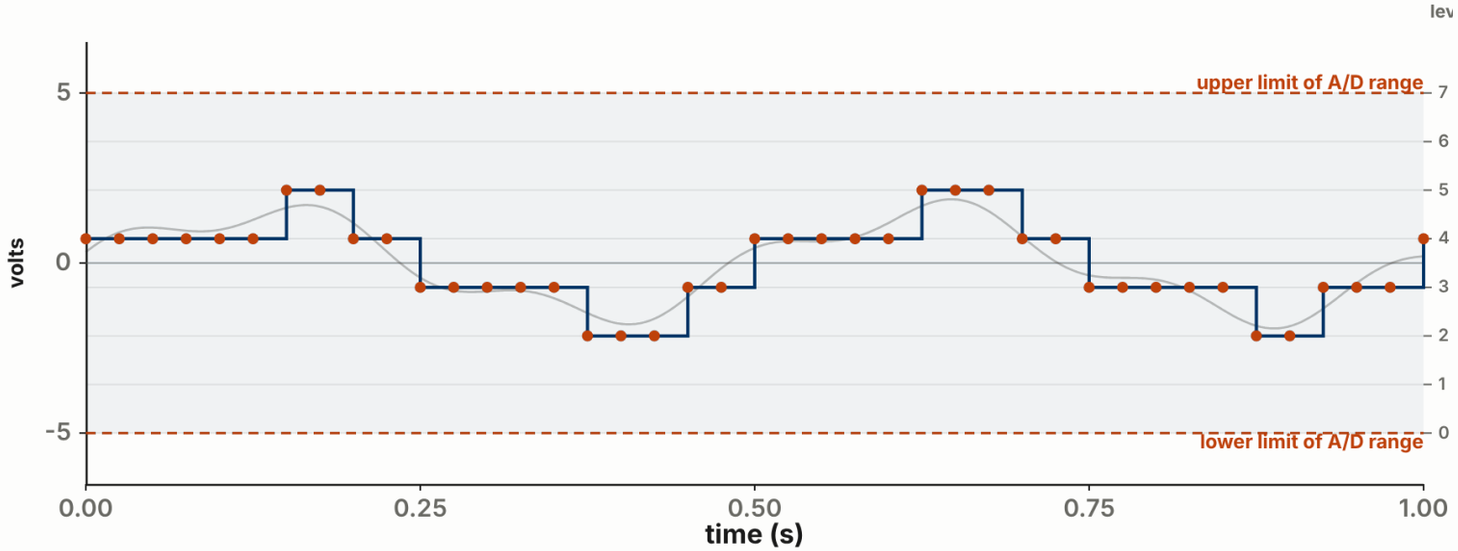
A common number of steps is **4 096**, meaning a given analog signal can be represented by any of 4 096 values between the limits of its range.

> For example, if the voltage range is 0 to 10 V, the amplitude between each step is
>
> $10 \,/\, 4\,095 = \mathbf{0.002\,44\ V}$

A high resolution is desirable because it allows each sample to represent the analog signal at that point in time **more precisely**.

Analog-to-digital converter

The amplifier facilitates sampling



range **-5** to **5** V · **3** bits = **8** levels, one step is **1.429** V · **40** Hz

only 8 levels to choose from, so the staircase is coarse — this is magnitude resolution

A/D RANGE **−5 to +5 V** | −10 to +10 V | 0 to 1 V

Magnitude — bits **3 bit → 8 levels**

Temporal — rate **40 Hz**

It does this by ensuring that the voltage input extends throughout **as much of the range of the A/D as possible**.

This increases the resolution of the sampled signal, by allowing the signal to be divided into more divisions.

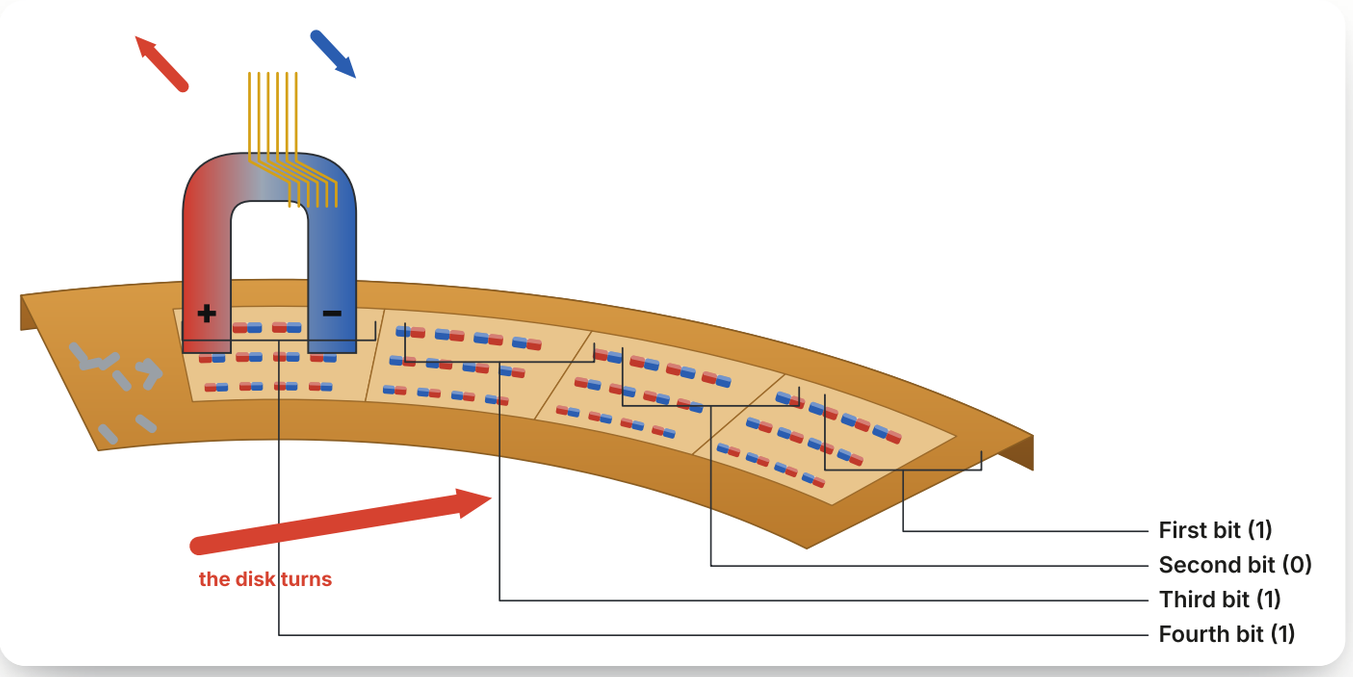
The ideal gain is one that **maximizes the range** of the A/D, but that **doesn't result in clipping**.

Data acquisition

Storage device

From the A/D, the digitized data — now in bits — are sent to a computer and stored in memory, such as a hard disk.

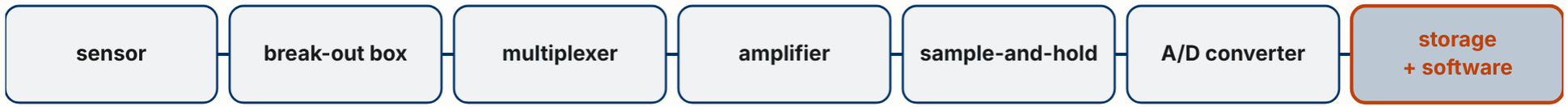
At this point the measurement is finished and everything that follows is arithmetic. Whatever the A/D threw away is gone for good: no amount of processing puts back a sample that was never taken, or a step that was too coarse.



Data acquisition

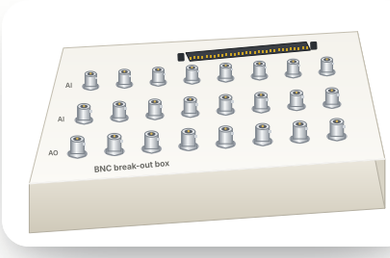
The whole chain

a physical quantity

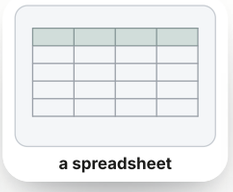


numbers on a disk

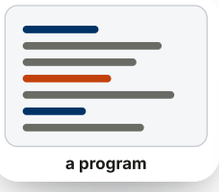
written down, then worked on



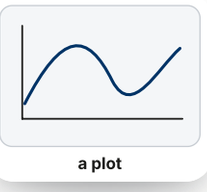




a spreadsheet



a program



a plot

Sensor → break-out box → multiplexer → amplifier → sample-and-hold → A/D → storage and analysis software.

Data acquisition

Software

The aim of any software package used to process and analyze data is to perform many operations on digital signals **very quickly**. Examples include separating the signal into channels and converting the values into physical units of measurement.

Often the user will have many operations to carry out on a set of data. Lists of these operations can be written in the form of a **program**, in one of many languages.

We may go over some data processing examples using various programs.

- Python
- R
- Matlab (Mathworks)
- LabView (National Instruments)
- Excel (Microsoft)



References

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- Griffiths, I.W., 2006. *Principles of Biomechanics & Motion Analysis*. Lippincott Williams & Wilkins, Baltimore, MD, USA; pp. 210–254.
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